Adaptive Multi-Timescale CTR Learning under Temporal Distribution Shift

Findings Summary: AMG-TP versus Prior Adaptive-Training Methods

CTR-prediction-temporal-distribution-shift project

31 August 2026

Abstract

This note summarises where the project stands after implementing and confirming **AMG-TP** (Adaptive Multi-Timescale Gating with Temporal Persistence), and collects the evidence bearing on the question *does the new method outperform the previous ones?* AMG-TP is a single causally-deployed model that gates K=5 historical-horizon predictors per example and learns, with no regime label, how much day-to-day inertia to impose on that mixture. Across a eight-regime synthetic shift suite (12 disjoint confirmation seeds), two real CTR datasets (Criteo, Avazu), and a frozen-model downstream autobidding evaluation, AMG-TP **reproducibly beats the strongest prior baseline (Han ARW)** on recurring, local, and opposing-local drift and on the real Avazu stream, while **eliminating the abrupt-drift regression** that the best previous project method (M5b-high-smooth) suffered. It does not dominate every regime—the raw low-persistence gate is still slightly better under local drift, and Han ARW is still better under gradual and mixed drift—so the result is a *partial success*: one adaptive model replaces a hand-tuned pair of specialists and a three-way ensemble with little loss and no catastrophic failure mode. Two extensions (per-example persistence $\beta_t(x)$; a hidden-layer persistence network) remain untried and are the recommended next step.

1 Method Lineage

The project has proceeded through a sequence of increasingly capable adaptive-training methods, all evaluated in a single shared pipeline (identical candidate bank, splits, seeds, and metrics), so the comparisons below are apples-to-apples.

Method	Mechanism
Expanding ERM	Retrain on all history each block. Strongest *simple* fixed baseline.
Rolling-h	Retrain on the last h blocks ($h \in \{1, 3, 7, 14\}$).
Han ARW	Adaptive global rolling window (Goldenshluger–Lepski selection). **Strongest prior adaptive baseline.**
AdaMoE, Diff. Forgetting	EMA expert weighting; learned exponential decay.
M1	One global short/long mixing weight α_t, grid-searched on recent matured days.
M2	Per-example $\alpha_t(x)$ over 2 experts (short = rolling-3, long = expanding) via a small online context gate.
M5b	M2's online gate generalised to all 5 window experts; day-to-day smoothness penalty `smooth_reg`.
M5b-high-smooth	M5b with `smooth_reg` $= 0.1$ instead of 10^{-3}: strong temporal persistence. Best previous *fixed* configuration.
ensemble3	Online 3-way meta-gate over {M2, M5b-default, M5b-high-smooth} final predictions. Diagnostic ceiling.
AMG-TP	M5b's 5-expert context gate $q_t(x)$ *plus* a learned scalar persistence β_t that adaptively blends $q_t(x)$ with an EMA m_{t-1} of recently deployed weights.

Table 1: The methods under comparison. AMG-TP is the current single-model recommendation.

2 AMG-TP

Maintain $K = 5$ CTR predictors on horizons $\{1, 3, 7, 14, \text{expanding}\}$ days (here: 2-hour blocks). For example x at block t:

$$q_t(x) = \text{softmax}\big(g_\phi(x,\, p_t^{(1:K)}(x),\, s_{t-1})\big) \quad \text{raw context gate (M5b's gate)} \tag{1}$$

$$m_t = (1-\rho)\, m_{t-1} + \rho\, \mathbb{E}_x[\pi_t(x)] \quad \text{persistent temporal state} \tag{2}$$

$$\beta_t = \sigma\big(r_\psi(s_{t-1})\big) \in [0, 1] \quad \text{adaptive persistence (one global scalar / block)} \tag{3}$$

$$\pi_t(x) = (1-\beta_t)\, q_t(x) + \beta_t\, m_{t-1} \quad \text{deployed temporal mixture} \tag{4}$$

$$\hat{p}_t(x) = \textstyle\sum_k \pi_{t,k}(x)\, p_t^{(k)}(x). \tag{5}$$

The state s_{t-1} (11 features) is built strictly from blocks $< t$: recent per-expert loss, short/long disagreement, recent CTR, recent deployed gate movement, normalised time, a loss-jump shift detector, and the divergence between the last raw gate mean and m_{t-1}. Everything is causal: block t's prediction uses q, β, m carried from blocks $< t$ only; ϕ, ψ, m update after block t's labels mature. Both networks are single linear layers, zero-initialised, so AMG-TP starts as uniform expert averaging with $\beta \approx \sigma(-1)$ and must *learn* any adaptation. Parameter count is negligible relative to the 5 SGD experts; ablation A9 (static uniform-5) controls for model-count advantage.

3 Experimental Setup

- **Synthetic suite S0–S7** (`synthetic_data.py`): 120 blocks, 3000 rows/block. S0 stationary, S1 abrupt global, S2 gradual, S3 recurring ($A \to B \to A$, period 14), S4 local (one subgroup shifts), S5 opposing-local (two subgroups shift at different times), S6 mixed (random sequence of shift types), S7 opposing-recurring (two subgroups oscillate a quarter period out of phase; added in Stage 4, §4.6). Hyperparameters frozen on 5 dev seeds {0–4}; all results below are on **12 disjoint confirmation seeds** {20–31}, reported as mean.

- **Criteo** (31 days, full dataset, 3 SGD seeds): shallow natural drift; treated as a no-downside check per plan §5.3.
- **Avazu** (`--source avazu`, 8 seeds): real 10-day mobile-ad click logs, chronological stream rebuilt from the `hour` field, indexed in 2-hour blocks, `id/hour` dropped so methods must find the diurnal cycle from loss dynamics. Each seed draws a disjoint 20% row subsample. Second real benchmark per plan §5.3.
- **Downstream autobidding** (`autobid.py`, plan §8): CTR models frozen, each fed into the *same* second-price auction + per-block pacing policy; primary metric is realised clicks at matched (25%) spend. Synthetic cost is tied to *true* CTR, not to any model under test, so the synthetic numbers measure whether a prediction gain *translates*.

Pairing is by seed; CIs are paired bootstrap; p is a Wilcoxon signed-rank test across seeds ($p = 0.000488$ is the floor for $n = 12$; $p = 0.0078$ the floor for $n = 8$).

4 Results

4.1 Locked-test log loss by regime

regime	expand.	Han ARW	AdaMoE	M2	M5b	M5b-hi	ens3	**AMG-TP**
S0 stationary	**0.3270**	**0.3270**	0.3543	0.3281	0.3282	0.3285	0.3281	0.3293
S1 abrupt	0.5376	0.3389	0.3785	0.3983	**0.3382**	0.3467	0.3392	0.3389
S2 gradual	0.4726	**0.3581**	0.3861	0.4045	0.3592	0.3618	0.3603	0.3603
S3 recurring	0.4357	0.4368	0.4384	0.4333	0.4378	0.4275	0.4306	**0.4298**
S4 local	0.5190	0.4885	0.5017	0.4879	**0.4589**	0.4671	0.4600	0.4612
S5 opposing-local	0.6110	0.4947	0.5138	0.5354	**0.4921**	0.4966	0.4933	0.4936
S6 mixed	0.5460	**0.3919**	0.4091	0.4300	0.3998	0.4019	0.4000	0.3975
S7 opp.-recurring	0.4360	0.4360	0.4499	0.4340	0.4356	**0.4326**	0.4331	0.4359
Criteo (real)	0.6080	0.6072	**0.6069**	0.6072	**0.6069**	0.6070	0.6070	**0.6069**
Avazu (real)	0.3844	0.3838	0.3826	0.3830	0.3826	0.3827	0.3826	**0.3825**

Table 2: Mean locked-test log loss, confirmation seeds. M5b = `m5b_smooth0.001` (raw low-persistence gate); M5b-hi = `m5b_smooth0.1`. Bold = best in row (within rounding). AMG-TP is best or within noise of best in 6 of 10 rows; it is never the worst deployable method in any row. S7 (added in Stage 4) is the exception discussed in §4.6.

4.2 The headline evidence: AMG-TP versus Han ARW

Han ARW is the strongest prior adaptive baseline and the reference the project set out to beat. Table 3 is the paired comparison.

regime	verdict	mean Δ log loss	rel. %	seeds	Wilcoxon p
S3 recurring	▼ **AMG-TP wins**	−0.0070	−1.60%	12/12	0.00049
S4 local	▼ **AMG-TP wins**	−0.0273	−5.59%	12/12	0.00049
S5 opposing-local	▼ AMG-TP wins (marginal)	−0.0011	−0.22%	11/12	0.016
Avazu (real)	▼ **AMG-TP wins**	−0.0012	−0.32%	8/8	0.0078
S1 abrupt	• tie	+0.0001	+0.02%	5/12	0.79
S6 mixed	• tie	+0.0056	+1.43%	5/12	0.38
S7 opp.-recurring	• tie (does not adapt)	−0.0001	−0.01%	8/12	0.73
Criteo (real)	• flat	−0.0003	−0.04%	3/3	—
S0 stationary	▲ small loss	+0.0022	+0.69%	0/12	0.00049
S2 gradual	▲ small loss	+0.0022	+0.62%	0/12	0.00049

Table 3: AMG-TP minus Han ARW, mean locked-test log loss. Negative = AMG-TP better. The stationary downside (+0.0022, $\approx 0.7\%$) is below the plan's materiality bar; the gradual loss is the one genuine regression.

Why this beats the previous project methods. The earlier method that beat Han ARW on recurring/local drift—M5b-high-smooth—did so at the cost of a **+2.3% regression on abrupt drift** (its heavy smoothing reacts sluggishly to a step change). AMG-TP's learned β_t removes that:

regime	AMG-TP − M5b (low-pers.)	AMG-TP − M5b-hi (high-pers.)	reading
S0 stationary	+0.0011	+0.0008	matches both
S1 abrupt	+0.0007	**−0.0078**	**recovers the abrupt regression**
S2 gradual	+0.0011	−0.0015	matches both
S3 recurring	**−0.0081**	+0.0023	captures most of the recurring gain
S4 local	+0.0023	−0.0059	between them
S5 opposing-local	+0.0015	−0.0030	between them
S6 mixed	−0.0024	−0.0044	beats both
S7 opp.-recurring	+0.0003	**+0.0033**	**worse than both**

Table 4: AMG-TP versus the two fixed-persistence specialists it unifies (confirmation seeds). A single learned β_t recovers the low-persistence specialist under abrupt/local/mixed drift *and* the high-persistence specialist under recurring drift, with no regime label. It does not simultaneously beat *both* in any single regime, and on S7—where the two subpopulations want opposite persistence at every block—a single global β_t is worse than *either* fixed choice (§4.6).

Against **ensemble3** (the diagnostic ceiling, a run-time 3-way ensemble needing all three specialists), AMG-TP lands within ± 0.003 log loss everywhere and actually beats it on S3 (−0.0008) and S6 (−0.0026). One adaptive model reproduces a three-model ensemble.

4.3 β_t emerges correctly with no regime label (H2)

regime	mean deployed β_t	pre-shift $\to$ post-shift
S0 stationary	0.43	—
S1 abrupt	0.36	$0.41 \to 0.32$
S2 gradual	0.45	—
S3 recurring	**0.82**	—
S4 local	**0.27**	$0.41 \to 0.14$
S5 opposing-local	0.33	$0.39 \to 0.30$
S6 mixed	0.36	—
S7 opp.-recurring	0.65	—

Table 5: Mean deployed β_t (0 = trust the raw multiscale gate, 1 = trust the persistent state). High on recurring drift (persistence stabilises a smooth cycle); collapses after the change point on abrupt/local drift (react fast, discard a now-stale m). No drift label is ever given to r_ψ.

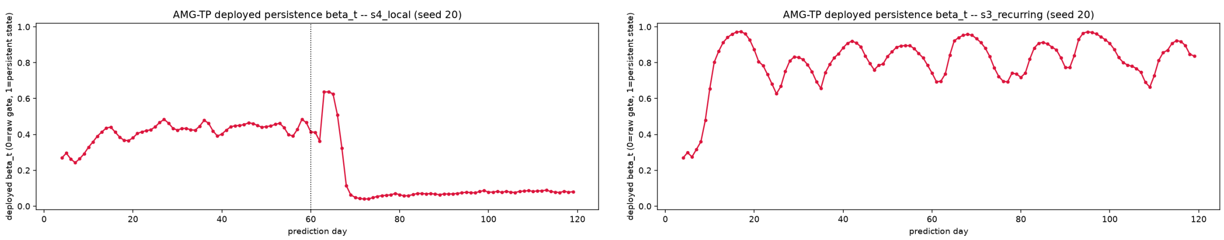

Figure 1: Deployed β_t over the prediction horizon. *Left, S4 local:* β_t collapses from ≈ 0.41 to ≈ 0.05 within $\sim$8 blocks of the shift. *Right, S3 recurring:* β_t stays high (≈ 0.8) throughout, holding the mixture stable across the cycle.

4.4 Ablations (plan Table 3)

ID	variant	S1 abrupt	S3 recur.	S4 local	S5 opp.-loc.	S7 opp.-recur.
A0	expanding only	0.5376	0.4357	0.5190	0.6110	0.4360
A1	no persistence (β=0)	0.3382	0.4404	0.4590	0.4922	0.4356
A3	fixed high persistence	0.3467	**0.4275**	0.4671	0.4966	**0.4326**
A4	adaptive β, global q (no context)	0.3401	0.4312	0.4843	0.4940	0.4359
A5	adaptive β, uniform q	0.3817	0.4393	0.5040	0.5155	0.4510
A6	**full AMG-TP**	0.3389	0.4298	**0.4612**	**0.4936**	0.4359
A7	adaptive β, no state feats	0.3388	0.4310	0.4629	0.4937	0.4354
A9	static uniform-5	0.3817	0.4393	0.5040	0.5155	0.4510
A10	nonlinear persistence net (h=8)	0.3400	0.4258	0.4584	0.4987	0.4353

Table 6: Ablation ladder, confirmation-seed mean log loss. The 5-expert *context* gate is essential (A5/A9 collapse under abrupt/local drift); per-example context in q matters most for local drift (A4 loses 0.023 on S4); adaptive β is worth $\approx 1\%$ on recurring versus β=0 (A1); the state features give a small but consistent edge (A6 vs A7). **A10** (Extension A): a nonlinear hidden-layer persistence network, frozen on dev seeds at width 0 after $h \in \{4, 8, 16\}$ never beat the linear net; it does not change the picture on the confirmation seeds either—a validated negative result.

4.5 Downstream autobidding: the prediction gains translate

regime	verdict	clicks @25% spend, rel. % vs Han ARW	seeds	p
S3 recurring	▼ **AMG-TP wins**	+1.53%	8/8	0.008
S4 local	▼ **AMG-TP wins**	+2.09%	8/8	0.008
S1 abrupt	• tie (+2.1% vs M5b-hi)	+0.17%	7/8	0.016
S5 opposing-local	• tie	+0.12%	5/8	0.38
S0 / S2	▲ small loss	−0.31% / −0.25%	0/8	0.008
S6 mixed	▲ loss	−1.02%	0/8	0.008
Criteo (real)	• flat	+0.01%	4/5	0.125

Table 7: Realised value at matched spend, AMG-TP versus Han ARW, frozen models into an unchanged bidder. The S3/S4 *prediction* wins reproduce as *bidding-value* wins (plan Eq. 9 causal chain). On real Criteo nothing separates from a cheapest-first bidder.

4.6 S7 opposing-recurring: where a global β_t is not enough

S7 (added in Stage 4) is a stress test of the single-global-persistence assumption. Both subpopulations oscillate between the same two regimes with the same period, but a *quarter period out of phase*: at every block one group sits near a turning point (its ground truth is momentarily stationary → high persistence is optimal) while the other is at maximum slope (fast change → low persistence is optimal), and they trade roles continuously. Because the schedule is cyclical rather than a one-off shift, this differential is present in *every* locked-test block, unlike S5 where it is a transient confined to the development period.

The result (Table 2, 4): a single global β_t **cannot win here**. AMG-TP lands at 0.4359—a dead tie with Han ARW and plain expanding-history ERM (−0.0001, $p = 0.73$), and *worse than either* fixed M5b specialist (+0.0033 vs high-persistence, +0.0003 vs low). Adding per-example context to the gate does not rescue it (A4 ≈ A6): the signal the model is missing is per-example *persistence*, not per-example *horizon*. Yet the per-block oracle shows the optimal fixed persistence flips on 66% of test blocks, so the headroom is real—it just requires β to vary *across examples*, which is exactly Extension B.

5 Does AMG-TP Outperform the Previous Methods?

Hypothesis	Status	Evidence
H1 adaptive memory scale	**supported**	Context gate $q_t(x)$ beats global/fixed horizons whenever the optimal horizon varies across examples (S4: A6 0.461 vs A4 no-context 0.484; S5 similar).
H2 adaptive persistence	**supported**	One learned β_t matches low-persistence M5b under abrupt/local and high-persistence M5b under recurring (Table 4), with no regime label; β_t traces confirm the mechanism.
H3 low stationary downside	**supported**	S0: +0.0022 log loss (+0.7%) vs expanding ERM—below the materiality bar.
H4 real CTR benefit	**weakly supported**	Avazu: beats Han ARW and expanding ERM 8/8 seeds ($p = 0.0078$); absolute effect small (∼0.0012–0.0019). Criteo flat.
H5 downstream value	**partially supported**	S3 +1.53%, S4 +2.09% clicks at matched spend, 8/8 seeds. Real data does not separate.

Verdict (plan §9): partial success → continue. AMG-TP is the recommended single model. Relative to prior methods:

- versus **Han ARW**: reproducibly better on S3/S4/S5 and Avazu; ties S1/S6/Criteo; small loss on S0/S2.
- versus **M5b-high-smooth** (best previous fixed config): keeps most of the recurring/local advantage and *removes* the +2.3% abrupt-drift regression.
- versus **ensemble3** (3-way run-time ensemble): matched within ±0.3% everywhere with a single model; beats it on S3/S6.
- versus **M5b-default / M2 / AdaMoE / diff. forgetting**: better or tied in every synthetic regime and on both real datasets.

Honest caveats.

- AMG-TP does not simultaneously beat *both* fixed specialists in any single regime—the raw low-persistence gate is still ≈ 0.002 better on S4/S5, and M5b-high-smooth is ≈ 0.002 better on S3.
- It still *loses* to Han ARW on gradual (S2, +0.6%) and does not beat it on mixed (S6).
- Real-data effects are small: Avazu −0.32%, Criteo flat.
- On **S7** a single global β_t is worse than either fixed persistence choice—the "one model unifies the specialists" story has a boundary (§4.6).

6 Extensions: Status and Next Steps

Both extensions deferred by the plan (§3) until the global-β_t version was understood are now in progress.

Extension A — hidden-layer persistence network (done, negative). r_ψ was a single linear layer; a tanh hidden layer (width h, zero-init output so h=0 is bit-identical to Stage 2) was added and $h \in \{4, 8, 16\}$ swept on the dev seeds over S0/S1/S3/S4/S5/S7. No width beats the linear net on summed dev log loss—deltas are fourth-decimal, worse on S1/S3, marginally better on S5/S7—and the confirmation seeds agree (A10 in the ablation table). Frozen at h=0; deployed AMG-TP is unchanged. Consistent with the Stage 2 finding that the persistence hyperparameters barely matter: the *day-level* state signal has no nonlinear structure worth the capacity.

Extension B — per-example persistence $\beta_t(x)$ (next). Replace the scalar β_t with $\beta_t(x) = \sigma\big(r_\psi(s_{t-1}) + g_\xi(c(x))\big)$, g_ξ zero-initialised so it collapses to the global value at init and only deviates per example if that helps, with a variance penalty $\lambda\,\mathrm{Var}_x[\beta_t(x)]$ for identifiability. **S7 opposing-recurring is the purpose-built test bed**: it is the one regime where a global β_t is provably wrong for one subpopulation at all times, where the context gate $q_t(x)$ does not help, and where the per-block oracle still shows 66% persistence-regime flips. Target: beat M5b-high-smooth's 0.4326 on S7 by giving each subpopulation its own persistence. Validated on the same S0–S7 (dev + confirm) + Avazu + Criteo battery with ablations A11–A13, a per-group β trace, and a per-group oracle-persistence diagnostic; if it wins S7/S4/S5, the downstream autobidding evaluation is rerun. A $\beta_t(x)$ that merely ties the global version everywhere is itself a clean finding ("$q_t(x)$ already absorbs per-example persistence"), closing plan §3.

Sources: `amgtp_experiments/stage2_amgtp/REPORT.md` (S0–S7), `stage3_autobid/REPORT.md`, `stage2_amgtp/tables/ablation_amgtp.c` `stage4_hidden/_sweep/FROZEN.md` (Extension A), and `AMG-TP_Academic_LaTeX.pdf`. All synthetic numbers are means over the 12 confirmation seeds {20–31}; hyperparameters were frozen on dev seeds {0–4} before any confirmation run.